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A lightweight decision-level fusion model for pig disease identification using multi-modal data

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ABSTRACT

In intensive pig farming, highly contagious diseases such as African swine fever, blue ear disease, transmissible gastroenteritis, and swine dysentery pose increasing threats to the pig industry. Timely detection of sick pigs and accurate diagnosis of their diseases are crucial for saving livestock. Therefore, this study integrates pigs' excretion behavior and fecal characteristics through a model decision layer and proposes the HCC-YOLOv8-LSTM model for pig health monitoring. First, the YOLOv8s model is improved by introducing the High Performance GPU Network (HGNet) module for feature extraction and fusion, which reduces computational costs and enhances feature extraction capabilities. The Cross-Stage Partial Convolution (CSPC) module, with Partial Convolution at its core, reduces the number of parameters and computations while effectively extracting spatial features, thereby enhancing the model's ability to identify abnormal pigs. Cross-Scale Feature Fusion (CSFF), as the Neck layer, fuses features of adjacent scales, enhancing the model's adaptability to health characteristics of pigs of different scales. Then, we introduce the Long Short-Term Memory (LSTM) network to extract time-continuous excretion behavior characteristics of pigs. By considering both excretion behavior and excretion characteristics through a decision layer fusion approach, the issue of poor accuracy due to single-mode features is reduced. The model achieves an mAP of 0.951 and a recall of 0.971. This lightweight model can be deployed on mobile devices, fits the practical conditions of pig farms, and can be applied to unmanned noncontact real-time detection systems to monitor pigs' health through their excretion behaviors and characteristics.

1. Introduction

With the rapid development of deep learning and multimodal fusion, smart agriculture is gaining increasing attention. In particular, pork is an important source of protein in many countries (Zhang et al., 2022b). Pig farming requires precise management strategies and continuous monitoring (Chen et al., 2021; Nie et al., 2023; Li et al., 2023). Therefore, pig disease monitoring is an indispensable part of smart agriculture and intelligent pig farming. Methods based on deep learning and multimodal fusion have stronger generalization capabilities and higher accuracy. By reducing the number of parameters and computational costs to achieve lightweight models, they can meet the needs of intelligent, noncontact disease identification and early warning systems in pig farms, gradually becoming the mainstream disease monitoring approach in pig farming.

Deep learning-based pig health monitoring models are generally divided into unimodal and multimodal models. Among unimodal models, convolutional neural networks (CNNs) have significantly improved health monitoring performance. With advancements in hardware performance and deep learning algorithms, various improved models have emerged, such as Faster Region-based Convolutional Networks (Faster R-CNN) (Ren et al., 2017), Vision Transformer (Dosovitskiy et al., 2020; Chen et al., 2023b), and You Only Look Once (YOLO) (Bochkovskiy et al., 2020; Ge et al., 2021; Li et al., 2022; Jocher and Chaurasia, 2023; Wang et al., 2024), achieving significant progress in performance and speed for object detection. These models have been effectively applied to pig health monitoring. For example, the DenseNet-121 model was improved using the Squeeze-and-Excitation Networks (SENet's) attention module to enhance the model's feature extraction

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capability, enabling accurate recognition of pig cough sounds for early warning of respiratory diseases (Song et al., 2022). A deep separable convolutional network was applied to classify pig postures, allowing for the early identification of factors that threaten pig health (Shao et al., 2021). The Mask Region-based Convolutional Neural Network (Mask R-CNN) model was improved by incorporating the Convolutional Block Attention Module (CBAM) and the Complete Intersection over Union (CIoU) loss function, allowing the model to focus better on the target areas of pig activities for automatic monitoring and early warning of abnormal activities in group-housed pigs (Yang et al., 2024).

In multimodal models, the main fusion strategies are divided into three types: data-level fusion, feature-level fusion, and decision-level fusion. With the increase in data and improvements in fusion algorithms, an increasing number of multimodal models are being applied to health monitoring. For instance, the YOLACTFusion model inputs RGB images and near-infrared images (900–1100 nm) into two ResNet50 backbone networks and uses a parallel attention mechanism to fuse feature maps of different scales into the head network, thereby enhancing the segmentation performance of the RGB image backbone (Liu et al., 2023). In the early warning of respiratory diseases in pigs, some studies have employed early fusion to integrate acoustic and deep features from audio segments for pig cough recognition (Shen et al., 2022) or combined acoustic and visual features to improve the accuracy of pig cough detection (Ji et al., 2022). Furthermore, corresponding heterogeneous deep features can be extracted from cough spectrograms and thermal images, with the spectrogram and thermal image features combined and fed into a support vector machine (SVM) classifier for cough classification (Wang et al., 2023). Additionally, classifier fusion strategies can be used, where SVM classifiers trained with acoustic and deep features are combined using soft voting for pig cough prediction (Yin et al., 2023).

In farming environments, pen hygiene has a significant effect on animal welfare and health (Nannoni et al., 2020). Failing to clean up feces promptly can lead to the accumulation of pathogenic microorganisms and parasites, thereby increasing the risk of diseases such as gastrointestinal disorders, urinary problems, mastitis, pleurisy, and lameness, which severely affect animal health and production efficiency (Van der Meer et al., 2016). The defecation behavior and fecal characteristics of pigs are closely related to their health status. Features such as the color and consistency of feces can provide valuable information for the early diagnosis of diseases. For example, if a pig experiences diarrhea, its feces may appear watery or viscous and could be accompanied by abnormal colors such as green or grayish-white, suggesting potential issues such as dysentery or intestinal infections. Abnormal components in feces, such as blood, pus, or parasitic eggs, often indicate potential disease infections, which affect the health of pigs. Additionally, abnormal defecation behaviors, such as frequent defecation or difficulty passing stools, may also be associated with systemic diseases. Noncontact monitoring of pigs, which is applicable to various environmental conditions, can operate for extended periods, and avoid the stress and safety risks associated with contact-based detection, improve production efficiency, animal welfare, and promote sustainable development. However, to the best of our knowledge, no studies on monitoring pig diseases through fecal analysis, particularly the use of deep learning models have been reported. This study thus fills this gap by providing a theoretical basis and foundation for noncontact early monitoring of pig diseases, laying the groundwork for precision health management in modern smart farming.

The study is organized into five sections, Section 1 presents the project background, industry status and existing problems. Section 2 describes the related work. Section 3 details the experimental data, setup, results, and analysis. Lastly, Sections 4 and 5 present discussions and conclusions on the experimental results obtained from the HCC-YOLOv8 model and HCC-YOLOv8-LSTM model.

2. Materials and methods

2.1. Data collection and image features

During the data collection phase, we collaborated with the College of Veterinary Medicine and four large-scale pig farms in Shanxi Province, each raising between 800 and 2500 pigs of the Long White Pig and Large White Pig breeds. This study focused on pigs in the nursery and fattening stages and collected information from pigs at different growth stages to increase the diversity of the dataset and improve the model’s generalizability. Our data collection period spanned from March 2023 to May 2024, during the data from a total of 56 healthy pigs, 13 pigs with African Swine Fever (ASF), 27 pigs with Porcine Reproductive and Respiratory Syndrome (PRRS), 22 pigs with Transmissible Gastroenteritis of Swine (TGE), and 16 pigs with Swine Dysentery (SD) were collected, ensuring the comprehensiveness and representativeness of the data. Abnormal behaviors or symptoms in pigs were monitored and detected by farm staff and veterinarians, who used biological detection methods to screen for ASF, PRRS, TGE, and SD. The samples were collected promptly and accurately to ensure high-quality data storage.

Video recordings were made using Hikvision DS-2CD 3345 D-I cameras, with a resolution of 2560x1440 pixels and a frame rate of 25 fps. The video data were stored in MP4 format. Cameras were placed at three corners of the pens, with two cameras at each position, at heights of 2.1 m and 1.3 m from the ground, respectively. This setup allowed for the collection of data on the excretion behaviors and excretions of healthy and abnormal pigs from different angles, ensuring data accuracy and diversity.

From monitoring to diagnosis and sampling, meticulous data collection methods laid the foundation for the pig excretion behavior dataset and the pig excretion dataset. Image annotation was performed using Labelimg, ensuring accurate labeling of excretion-related features.

To meet the model’s training needs and improve its generalization ability, several data augmentation techniques, such as Horizontal and Vertical Flipping, Brightness Adjustment, and Random Cropping, have been employed. These techniques increase data diversity and ensure that the model trained on augmented data can be generalized better to new, unseen examples. A comparison of the datasets before and after data augmentation is shown in Table 1.

When monitoring the health of pigs, it is crucial to capture the excretion behavior characteristics and excretion characteristics caused by diseases. Table 2 shows the relevant excretion behaviors and fecal characteristics of pigs with different diseases and healthy pigs, including African Swine Fever (Fig. 1(a)), Porcine Reproductive and Respiratory Syndrome (Fig. 1(b)), Transmissible Gastroenteritis of Swine (Fig. 1(c)), Swine Dysentery (Fig. 1(d)), and healthy pigs (Fig. 1(e)).

2.2. YOLOv8s model

YOLOv8 (Jocher and Chaurasia, 2023) is an efficient single-stage object detection algorithm that extends the network architecture

Table 1
Comparison of datasets before and after data enhancement.

Dataset	Classes	Number(original dataset)	Number(augmented dataset)
Excretion behavior	Normal	440	1760
	ASF	226	904
	PRRS	281	1124
	TGE	263	1052
	SD	208	832
Manure	Normal	630	2520
	ASF	343	1372
	PRRS	401	1604
	TGE	366	1464
	SD	321	1284

Table 2

Fecal characteristics of pigs with different diseases.

Condition	Excretion behavior features	Fecal features
African swine fever	(i)The defecation shows marked distress, with frequent rising and falling and stirring. (ii)Sometimes a cry of agony.	(i)Excrement with blood or dark red color, accompanied by mucus. (ii) May contain partially digested food particles.
Porcine reproductive and respiratory syndrome	(i)Defecation may be halting, is often followed by coughing or deep breathing. (ii) Occasionally shows transient weakness or discomfort.	(i)Excrement are relatively soft, without blood, and may appear light yellow or green. (ii) There will be no obvious mucus or blood.
Transmissible gastroenteritis of swine	(i)The frequency of defecation increased significantly. (ii) When defecating he appeared very weak and frequently lay down.	(i)Excrement appear watery, with undigested food particles. (ii) Usually yellow or green in color, and without blood.
Swine dysentery	(i)Show extreme discomfort when defecating, with frequent rising and falling. (ii) Scratching of the stomach, and possibly even twitching or trembling.	(i)The Excrement are watery and contain a large amount of mucus and blood.(ii) The Excrement may be red, black, or mixed in color, and often contain undigested food particles. (iii) In the initial stage, yellow to gray soft stool is discharged. (iv) When dysentery persists, mucus, blood, and cellulose fragments can be seen mixed in the stool, making it brown, red, or black.
Normal	No distinctive features	No distinctive features

concepts of YOLOv5, including the Cross Stage Partial Network based (CSP-based) Backbone and Neck design, as well as models of different scales on the basis of scaling factors. YOLOv8 introduces a new state-of-the-art (SOTA) model, featuring object detection networks with resolutions of P5 640 and P6 1280. The Backbone and Neck sections replace the C3 structure of YOLOv5 with the gradient-rich C2f structure, adjusting the number of channels for different scale models, and significantly improving model performance. The Head section separates the classification and detection heads and shifts from Anchor-Based to Anchor-Free, enhancing the model's flexibility. The output maps are processed through the Spatial Pyramid Pooling – Fast (SPPF) module, which supports different resolutions and ratios. For loss calculation, the Task-Aligned Assigner positive sample allocation strategy and Distribution Focal Loss are used for effective computation, improving the model's stability and generalization ability. In the training data augmentation phase, the last 10 epochs of Mosaic augmentation are turned off, as in YOLOX, further enhancing the performance of YOLOv8 object detection.

2.3. K-fold cross-validation

This study introduces the K-fold cross-validation method for dividing the dataset. Specifically, the dataset is divided into K mutually exclusive subsets of similar size, called “folds.” The model is trained and evaluated K times, with each iteration using K-1 folds as the training set and the remaining fold as the validation set. The validation fold was changed in each iteration to better represent the dataset. Additionally, this method is relatively mature and widely used in machine learning (Zhang and Liu, 2023), as it enables more effective utilization of the data, improving the stability of evaluation results and reducing bias caused by different data partitions. A diagram of K-fold cross-validation is shown in Fig. 2.

2.4. HCC-YOLOv8 model

The HCC-YOLOv8 model primarily consists of three components: the backbone, neck, and head. In the backbone, we reference the real-time detection model RT-DETR (Zhao et al., 2024) based on Transformer and replace the YOLOv8 backbone with its core component, HGNet, achieving a lightweight design while maintaining high accuracy and computational efficiency. In the neck layer, we introduce the CSFF module (Zhao et al., 2024) to modify the original structure of YOLOv8; this module fuses features from two adjacent scales to improve the model's accuracy in recognizing pig manure at different scales and enhance the model's generalization ability. Additionally, we propose the CSPC module, centered on Partial Convolution (PConv) (Chen et al., 2023a), to replace C2f in YOLOv8 with a lightweight design. For the head, we apply the Self-Attention mechanism from the SOTA segmentation model HyCTAS (Yu et al., 2024) to the detection head, forming a self-attention detection head that reduces the feature dimensions and computational load. Furthermore, the EIoU loss function (Zhang et al., 2022a) is incorporated to further improve model accuracy. The overall structure of HCC-YOLOv8 is shown in Fig. 3. HCC-YOLOv8 significantly improves the accuracy of detecting feces from African Swine Fever (ASF), Porcine Reproductive and Respiratory Syndrome (PRRS), Transmissible Gastroenteritis of Swine (TGE), and Swine Dysentery (SD), providing a reliable early monitoring and warning tool for the pig farming industry.

2.4.1. HGNet module

To address the challenges faced by Vision Transformers (Li et al., 2023), such as low memory efficiency, heavy computational costs, and unsuitability for real-time applications, Baidu introduced the new real-time ViT model RT-DETR. HGNet is the backbone of RT-DETR, which focuses on efficient feature extraction within constrained computational resources. It employs a lightweight design, optimizes convolutional operations and leverages depthwise separable convolutions to reduce computational complexity. Additionally, HGNet incorporates a Learnable Down-Sampling (LDS) layer to perform downsampling, effectively reducing the spatial dimensions of feature maps and alleviating computational overhead. By integrated convolution and attention mechanisms, HGNet combines local feature extraction with global contextual information, enhancing its feature representation capabilities. The network constructed multi-scale feature representations to efficiently detect objects of varying sizes and dynamically adjusted the receptive field to accommodate multi-scale detection tasks. To further improve robustness against spatial transformations in input data, HGNet uses a Global Average Pooling (GAP) layer before the final classification, reducing the spatial dimensions of feature maps to a single vector per feature map. The network culminates with a series of layers designed for final classification, typically involving a convolutional layer for feature aggregation followed by a fully connected layer to map these features to the desired output classes. This hierarchical approach to feature extraction enhances efficiency and aligns with the need for lightweight architectures.

In light of these advantages, HGNet replaces the YOLOv8 backbone to meet the lightweight requirements of pig farming operations, ensuring an optimized solution for real-world applications in resource-constrained environments.

2.4.2. CSPC (Cross-Stage Partial Convolution) lightweight module

The core of the CSPC Lightweight Module is Partial Convolution (PConv) (Chen et al., 2023a), which replaces the C2f module in the model. This improvement eliminates highly similar channel features, utilizes feature map redundancy, and reduces computational redundancy and memory access, significantly lowering the number of parameters and achieving lightweight detection heads, thus increasing the detection speed. Fig. 6 illustrates how our PConv operates. Specifically, PConv applies conventional convolution only to a portion of the input



(a) African Swine Fever



(b) Porcine Reproductive and Respiratory Syndrome



(c) Transmissible gastroenteritis of swine

Fig. 1. Sample images of pig excretion behavior and fecal characteristics.



(d) Swine dysentery



(e) Normal

Fig. 1. (continued).

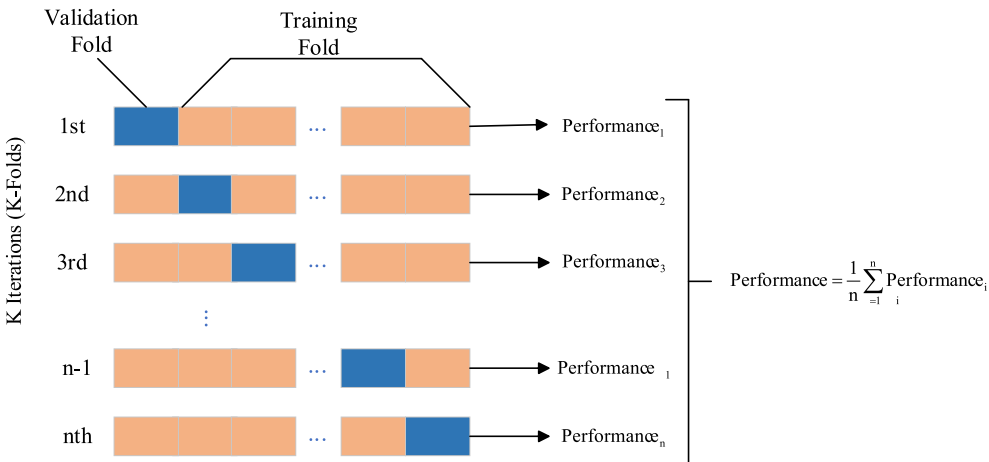
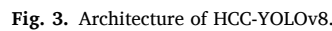


Fig. 2. K-fold cross-validation.



The HCC-YOLOv8 model achieves satisfactory detection results for

image data feature recognition. However, when dealing with video stream data that has temporal continuity, the model fails to analyze the temporal information in the data. The LSTM network structure includes a feedback mechanism in its hidden layers, which allows the current moment's training results to consider both the input data features of the current moment and the output results from the previous moment, making it suitable for detecting anomalies in pig defecation behavior with temporal features. Therefore, features obtained from the HCC-YOLOv8 network are fed into the first LSTM layer in the order of video sequence, and the output of the LSTM layer is used as the input for the next LSTM layer. Using a two-layer LSTM network improves the recognition of abnormal defecation behavior.

We use the HCC-YOLOv8-LSTM network for detecting anomalies in pig defecation behavior and identifying pig feces. These results contribute to the early detection of various pig diseases, including African Swine Fever (ASF), Porcine Reproductive and Respiratory Syndrome (PRRS), Transmissible Gastroenteritis of Swine (TGE), and Swine Dysentery (SD). By monitoring the changes in defecation behavior and feces characteristics, our approach aims to identify signs of these diseases at an early stage. The overall workflow is illustrated in Fig. 4.

First, we determined whether the pig exhibited abnormal defecation behavior. If no abnormal behavior is detected, the feces identification result is used for ASF detection. If abnormal defecation behavior is detected, the results of abnormal behavior detection and feces identification are combined using weighted summation to produce the final ASF detection result. The weighted summation formula is as follows:

$$FinalResult = \omega_1 Result_e + \omega_2 Result_f \quad (6)$$

where, $Result_e$ represents the detection result of abnormal pig defecation behavior and $Result_f$ represents the identification result of pig feces. ω_1

and ω_2 are the weights, which are set to 0.35 and 0.65, respectively, on the basis of our testing and recommendations from domain experts.

2.6. Training settings and performance indicators

2.6.1. Training settings

The experiments were conducted on a system with a 2.50 GHz CPU and 32 GB of RAM, providing sufficient computational power and memory for model training while meeting the lightweight deployment requirements for pig farms. The software environment included Windows 11 for its stability and widespread use in farm operations, with Python 3.9.2 serving as the programming language owing to its extensive scientific libraries for machine learning and data analysis. Deep learning tasks were implemented using PyTorch 2.0.1, whereas Numpy 1.26.4 and Pandas 2.2.2 were used for numerical processing and data manipulation. Additionally, OpenCV 4.10.0 facilitated image processing, and Scikit-learn 1.4.2 supported model evaluation and other machine learning tasks.

The training process uses Adaptive Moment Estimation (Adam) (Kingma and Ba, 2015) as the optimizer, which integrates the principles of momentum and RMSProp to adjust the learning rate for each parameter through first-order and second-order moment estimates of the gradients. This approach facilitates more efficient training. The initial learning rate is set at 0.01 and decreases via Step Decay, which is reduced by a factor of 10 every 50 epochs to ensure convergence. The batch size is set to 8, and the total number of training epochs is 300. These hyperparameters were chosen on the basis of preliminary experiments and fine-tuned according to the model's validation performance.

2.6.2. Performance indicators

In the study of pig disease detection using a multimodal fusion

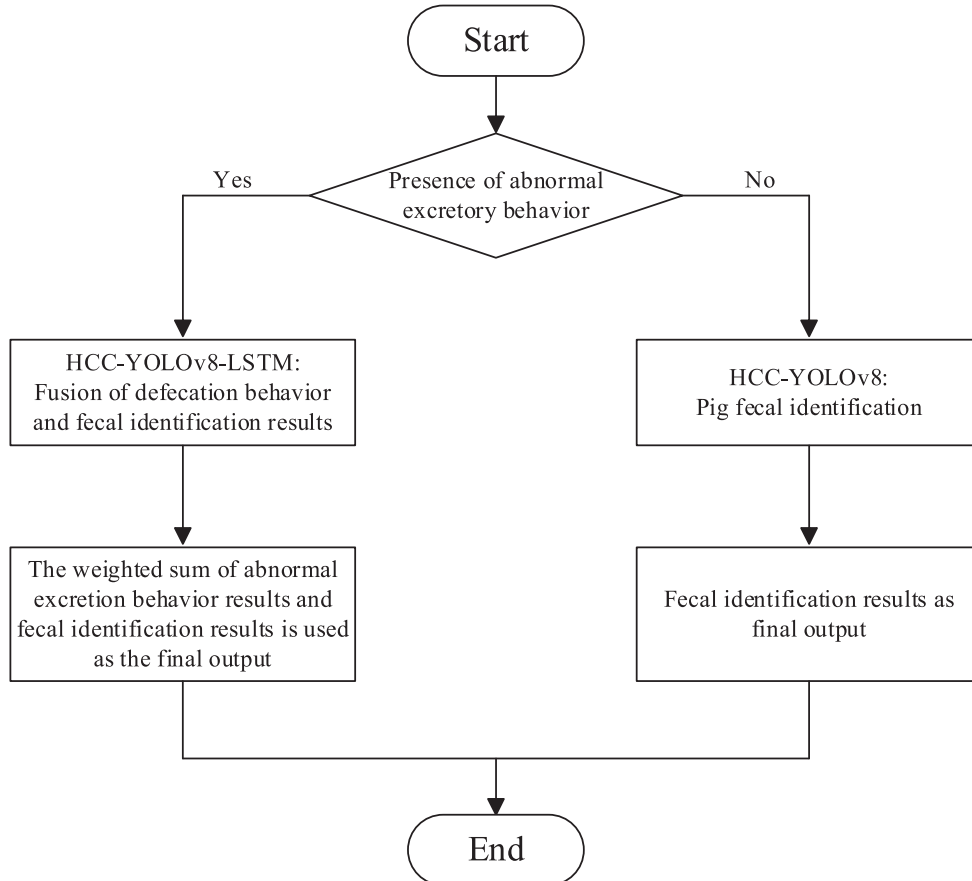


Fig. 4. Overall flowchart of HCC-YOLOv8-LSTM.

recognition neural network model, evaluating the model's effectiveness is crucial. We employed a set of selected metrics to assess the model's performance and lightweight characteristics.

Model accuracy is a commonly used metric that represents the ratio of correctly predicted positive samples to the total number of predicted positive samples. Recall measures the ratio of correctly predicted positive samples to the total actual positive samples, reflecting the model's ability to capture true positives. The F1 score harmonizes precision and recall, demonstrating a balance between the model's accurate detection and comprehensive coverage. Given the inverse relationship between precision and recall, achieving high levels of both metrics is challenging. Therefore, the mean Average Precision (mAP) becomes prominent. The mAP is calculated by plotting the Precision-Recall (PR) curve, with precision on the y-axis and recall on the x-axis. The area under the curve, enclosed by the curve and the axes, represents the Average Precision (AP) for each category. The average of these AP values across all categories provides the mean Average Precision (mAP).

The metrics used to measure the lightweight characteristics of the model are Params and GFLOPs. Params refers to the total number of parameters in the model, including weights and biases. GFLOPs represents the billions of floating-point operations performed per second, and its size is used to measure the computational load during forward propagation. For a lightweight model, the requirements for Params and GFLOPs are to be as low as possible while maintaining performance.

3. Experimental settings

3.1. Single characteristics for pig health monitoring

3.1.1. HCC-YOLOv8 model results

In this study, we utilized the HCC-YOLOv8 model to detect differences in the feces of pigs with various diseases, and explored the feasibility of using the improved YOLOv8 model for pig disease detection. Fig. 5 shows the average loss during model training and validation, illustrating the dynamic balance between this loss and the model's capabilities in bounding box accuracy and object detection. The decrease in average loss indicates the improved accuracy of bounding box localization and the effectiveness of feces recognition with the enhanced model. During the training phase, the model underwent 300 iterations.

The train/box_loss and val/box_loss show the loss for training and validation over epochs, respectively; in these graphs, the loss decreases and stabilizes over epochs. From train/box_loss, it is evident that the loss decreases from 3.01 to 0.68. This reflects the model's flexibility in identifying key fecal features indicative of different diseases. The cls_loss indicates the difference between the predicted and true classes. After 300 iterations, it stabilized at approximately 0.70, demonstrating the model's accuracy in detecting subtle features of feces from pigs with different diseases. As a custom loss function introduced in YOLOv8, dfl_loss is used to compute the difference between the predicted feature points and ground truth feature points, aiding in predicting object orientation and angle information. The train/dfl_loss results clearly show that the model effectively captures feature points of feces from pigs with various diseases, supporting its ability to distinguish between different disease conditions. For the validation set, the loss function generally shows a decreasing trend, with slight fluctuations, but ultimately stabilizes. The validation metrics demonstrate the model's robust performance. The metrics/precision curve establishes the model's precision in identifying feces from pigs with different diseases. The metrics/recall curve reveals the model's proficiency across various IoU thresholds. Additionally, metrics/mAP50 rapidly increased to 0.86 after the model's initialization and gradually stabilized at approximately 0.93. On the other hand, metrics/mAP50-95 provided finer insights, starting from an initial value and steadily increasing to 0.77, highlighting the model's generalizability across different IoU evaluations.

3.1.2. Confusion matrix analysis

A confusion matrix is a commonly used metric in deep learning for evaluating classification models. It represents the relationship between the predicted and true classes in matrix form. The rows of the matrix indicate the true classes, whereas the columns represent the predicted classes. The values on the diagonal of the matrix represent the accuracy of correct classifications made by the model, whereas values off the diagonal indicate misclassification rates. In feces recognition tasks, the confusion matrix helps understand the model's performance across different categories and identify potential areas of error. Analyzing the confusion matrix can reveal the model's weaknesses and guide targeted improvements.

Fig. 6 shows the confusion matrix of the HCC-YOLOv8 model in the

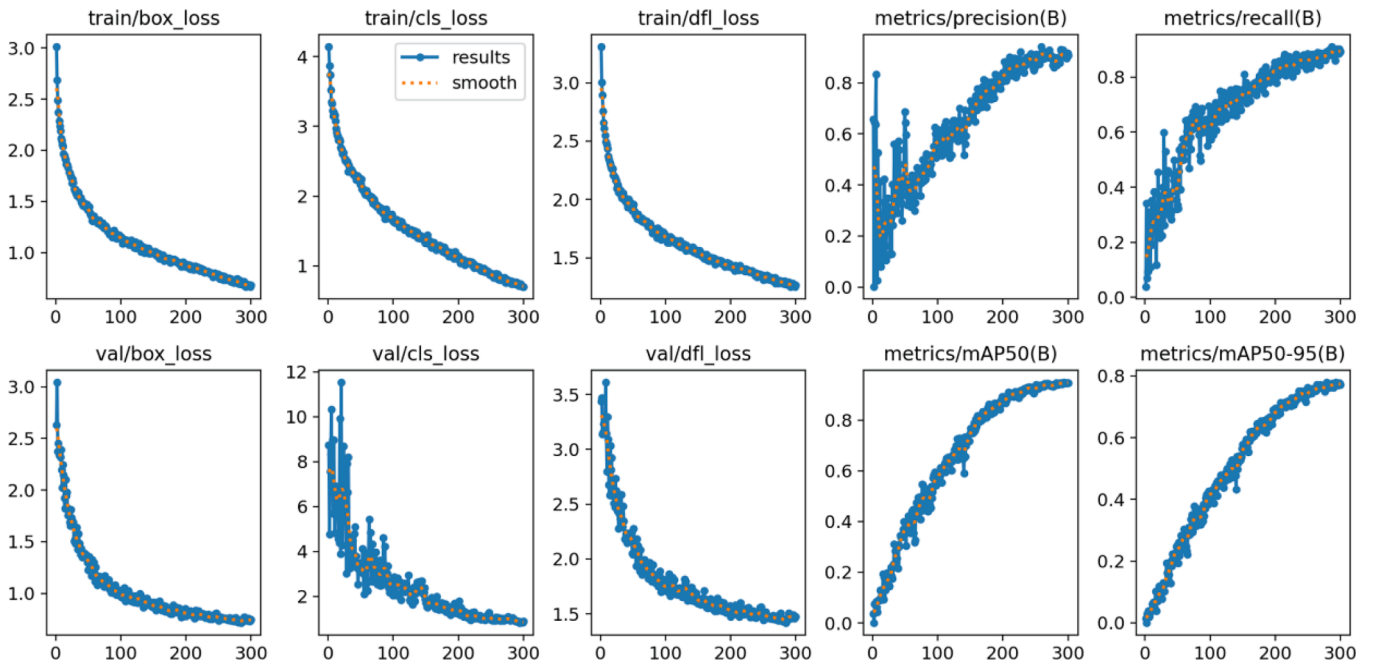


Fig. 5. Identification results of pig feces using HCC-YOLOv8.

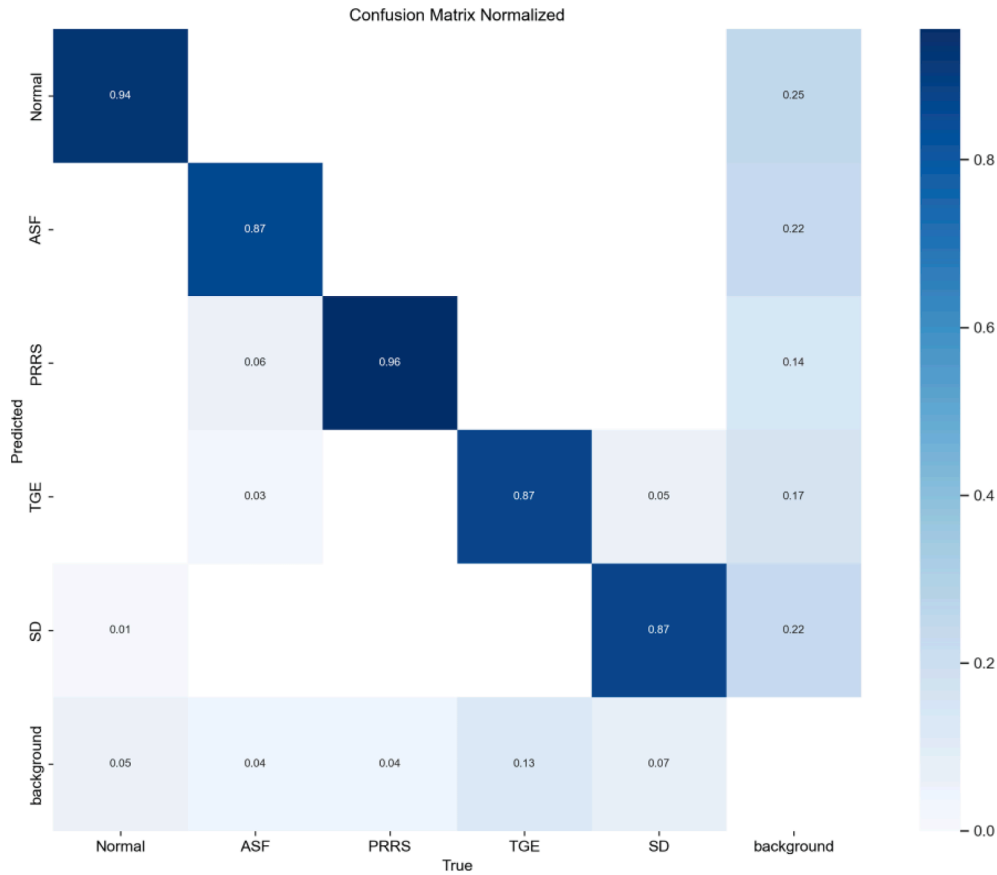


Fig. 6. Confusion matrix of HCC-YOLOv8 in fecal recognition.

task of disease recognition based on pig feces. The model performed well across five different categories: “Normal”, “ASF”, “PRRS”, “TGE”, and “SD”. The model achieved an accuracy of 0.94 for identifying healthy pigs’ feces (Normal). While high accuracy in detecting healthy samples contributes to overall model performance, the primary focus remains on the early detection of pig diseases. Additionally, achieving a balance between minimizing false-positives and maximizing sensitivity is crucial in disease detection, as an appropriate level of false-positives can indeed enhance early detection. This allows the model to effectively prevent potential outbreaks by identifying disease cases with greater sensitivity, even at the cost of occasional false alarms. For ASF-affected pigs’ feces, the model exhibited excellent performance with an accuracy of 0.93, allowing the precise identification of African Swine Fever. The model also performed well for PRRS with an accuracy of 0.93; however, there was significant confusion with ASF, resulting in a misclassification rate of 0.06, which could lead to incorrect diagnoses. The model predicted TGE with an accuracy of 0.91, but there was some confusion with SD, with a misclassification rate of 0.07. This confusion might be due to the similarity in fecal features of pigs with these two diseases. The model had the lowest accuracy for SD, at 0.82. This may be attributed to the fact that most of the collected data were images of early symptoms of SD, with a relatively smaller dataset. Further optimization is needed to improve differentiation. Despite these challenges, the model demonstrates reliable performance in distinguishing early symptoms of SD. The achieved accuracy indicates that the model has potential for further enhancement through data expansion and optimization of feature extraction for complex patterns.

In summary, the HCC-YOLOv8 model proposed in this study performed excellently overall in the task of pig health monitoring, achieving a high accuracy rate. However, there is still some confusion among certain categories, leading to significant misclassifications. These category characteristics need further analysis to improve the model and

enhance its distinguishing capabilities. Additionally, careful consideration should be given to balancing samples among categories to prevent the model from being overly biased toward specific categories. While the current model performance is commendable, there is still room for improvement.

3.1.3. Ablation experiments

The purpose of ablation experiments on model components is to verify the impact of improvements in each part of the model on overall performance. Using YOLOv8s as the baseline, we conducted a series of ablation experiments on the three network components mentioned in Section 3.1 (shown in Fig. 3). The performances of the different components are shown in Table 3. Replacing the original CioU in YOLOv8 with our selected EIoU improved the model’s recognition accuracy without increasing computational costs, with mAP50 and mAP increasing by 0.8 % and 1.3 %, respectively. When HGNet was used as the backbone, the mAP50 improved by 3.0 % and the mAP improved by 2.9 % compared with those of the baseline, whereas the number of parameters decreased by 2.8 M, and the computational cost decreased by 4.3 GFLOPs. This finding demonstrates that HGNet is beneficial for extracting multi-scale features and can help in lightweighting the model to some extent. Compared with the baseline, when only the CSFF module was used to improve the neck layer, there was a slight improvement in mAP50 and mAP, with a reduction of 1.3 M in the number of parameters and 2.2 GFLOPs in computational cost. This finding indicates that changes to the neck layer contribute to the lightweighting of the model. Replacing the C2f module with the CSPC module further achieves model lightweighting, although it results in a slight reduction in accuracy. However, it significantly reduces the number of parameters and computational cost, greatly promoting the model’s lightweighting. Introducing the Self Attention mechanism in the head layer improved the recognition accuracy of multi-scale targets.

Table 3

Ablation experiment of model components. “SA” represents Self Attention module.

EIoU	HGNet	CSFF	CSPC	SA	mAP50	mAP	Params	GFLOPs
✓					0.887	0.694	11.1 M	28.4
✓					0.895	0.707	11.1 M	28.4
✓	✓				0.917	0.723	8.3 M	24.1
✓		✓			0.911	0.715	9.8 M	26.2
✓			✓		0.891	0.703	7.6 M	19.8
✓				✓	0.913	0.719	10.8 M	25.9
✓	✓	✓			0.921	0.739	10.5 M	25.4
✓	✓		✓		0.914	0.711	6.6 M	20.6
✓	✓			✓	0.922	0.740	7.9 M	21.4
✓	✓	✓	✓		0.919	0.721	8.7 M	25.3
✓	✓	✓		✓	0.933	0.754	9.2 M	26.1
✓	✓	✓	✓	✓	0.920	0.736	7.3 M	18.4
✓	✓	✓	✓	✓	0.927	0.748	3.8 M	14.4

We subsequently combined different model components to verify the impact of these combined improvements on model performance, as shown in Table 3. Among all the combinations, HCC-YOLOv8 achieved the second-highest recognition accuracy with the lowest parameter count and computational load. Considering the hardware requirements for deploying the model in real pig farms, we selected it as the improved model for our experiments.

3.1.4. Comparison experiments

The results of the different models for pig feces recognition are summarized in Table 4. All models were trained from scratch without using pretrained models. HCC-YOLOv8 achieved impressive recognition performance with the lowest parameter count and computational load, reaching 92.7 % mAP50 and 74.8 % mAP. Compared with other lightweight models and YOLO series models, it achieves state-of-the-art performance. The recognition results of HCC-YOLOv8 are shown in Fig. 7.

ShuffleNetV2, EfficientNetV2 (Tan and Le, 2021), and MobileNetV3 (Howard et al., 2019) are popular lightweight networks. Compared with ShuffleNetV2, HCC-YOLOv8 achieves a 15.9 % greater mAP50 and a 21.4 % greater mAP with $\times 2$ inference speed (200 FPS vs. 122 FPS). Compared with EfficientNetV2, our model improves the mAP50 and mAP by 10.7 % and 12.9 %, respectively, while significantly reducing the parameter count. Although MobileNetV3 has superior lightweight performance, our model significantly outperforms it in accuracy and achieves satisfactory results in terms of parameter count and computational load.

In addition, this experiment compares the performance of HCC-YOLOv8 with that of other prominent object detection models, such as YOLOv5, YOLOv7, YOLOv8, and YOLOv10, focusing specifically on the s-size variants where available. Each model was independently trained to converge and then evaluated on the basis of its ability to distinguish diseases from pig feces. Compared with these other models, HCC-YOLOv8 achieves a better balance between speed and accuracy. Compared with YOLOv5s, YOLOv7, and YOLOv8s, the HCC-YOLOv8 model demonstrates a greater degree of lightweight performance, with a substantial reduction in both parameters and the computational load.

Table 4

Comparisons of object detectors.

Model	Input size	mAP50	mAP	Speed (FPS)	Params	GFLOPs
ShuffleNetV2	640 $\times$	0.768	0.534	122	15.4 M	59.7
EfficientNetV2	640	0.822	0.619	115	24.1 M	30.2
MobileNetV3		0.804	0.586	156	12.4 M	21.9
YOLOv5s		0.865	0.668	121	29.5 M	49.1
YOLOv7		0.872	0.683	126	26.3 M	38.9
YOLOv8s		0.887	0.697	109	24.2 M	28.6
YOLOv10s		0.924	0.741	178	8.8 M	24.2
HCC-YOLOv8		0.927	0.748	232	3.8 M	14.4

Improvements in multi-scale feature extraction and fusion have also led to satisfactory results in terms of recognition accuracy. Compared with the recently proposed YOLOv10s, our model has a lower computational load and higher accuracy, with mAP50 and mAP improved by 0.3 % and 0.7 %, respectively.

In summary, HCC-YOLOv8 has significant advantages in terms of model size, speed, and recognition accuracy. This method effectively meets the requirements for the noncontact differentiation of different diseases through feces in the real world and achieves a lightweight model design that meets the practical needs of pig farms.

3.2. Evaluation of the results of the HCC-YOLOv8-LSTM

3.2.1. Evaluation of decision-level fusion

We evaluated the performance of the fusion model and compared it with the models before fusion, as well as with models obtained by using different weights for fusion, as shown in Fig. 8. In the figure, Fusion_1 (0.2, 0.8) represents the fusion of the output results from HCC-YOLOv8 and LSTM with weights of 0.2 and 0.8, respectively. The weights refer to the parameters in Equation (6). After the output results were fused, the evaluation metrics were calculated on the basis of the number of positive and negative samples. The meanings of Fusion_2 (0.3, 0.7), Fusion_3 (0.4, 0.6), and Fusion_4 (0.5, 0.5) are similar. The weight parameters used for HCC-YOLOv8-LSTM are 0.35 and 0.65, which are mentioned in the explanation of Equation (6) and were determined on the basis of experimental results and expert recommendations.

The figure shows that when using HCC-YOLOv8 or LSTM alone, there is some error in the recognition accuracy, which is not conducive to accurate detection and early warning of pig diseases. Moreover, the performance of the fusion model varies with different weight parameters. Reasonable selection of weights can effectively improve the accuracy of disease detection in pigs, providing support for early monitoring of pig health.

3.2.2. Comparison experiments

This section compares the fusion results with single-modal recognition results to verify the reliability of integrating pigs' excretion behaviors with excretion characteristics. The YOLO-DNA model, which is used for comparison, is a lightweight multimodal model that is based on YOLOv7-tiny and employs an MLP-based multimodal fusion encoder (Chen et al., 2024). Fig. 9(a) shows a comparison of the recognition accuracy. Overall, the fusion model achieves higher recognition accuracy than the single-modal models do and exhibits relatively smaller fluctuations during training. This finding indicates that integrating both pigs' excretion behaviors and excretion characteristics results in more accurate disease recognition, such as African swine fever, and reduces misclassifications associated with single-modal data. Fig. 9(b) displays the loss comparison results. The loss of the fusion model decreases more rapidly, and our model achieves a lower loss value, indicating that our model better extracts key features and has smaller differences between

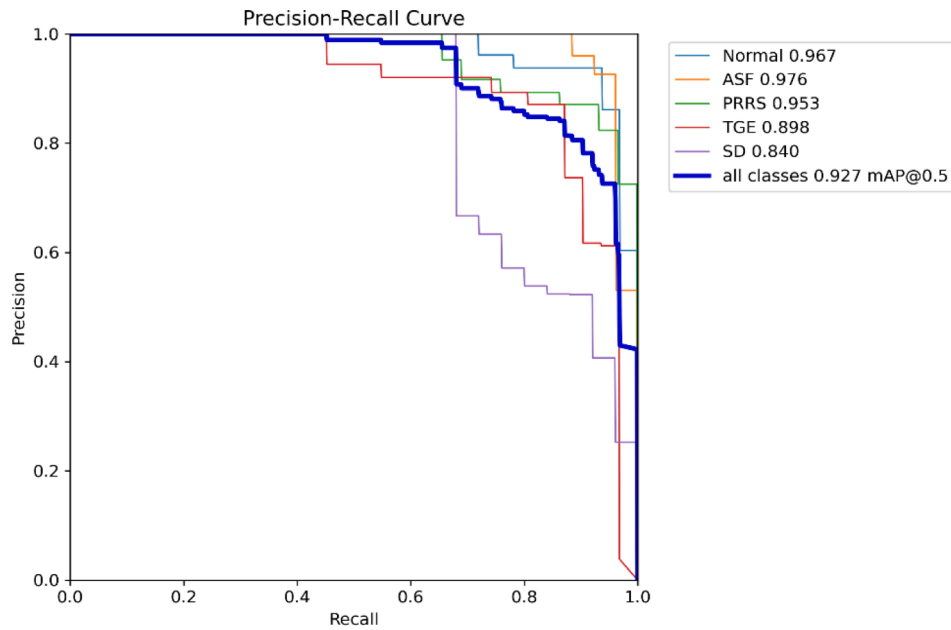


Fig. 7. The mAP50 of HCC-YOLOv8 in fecal recognition.

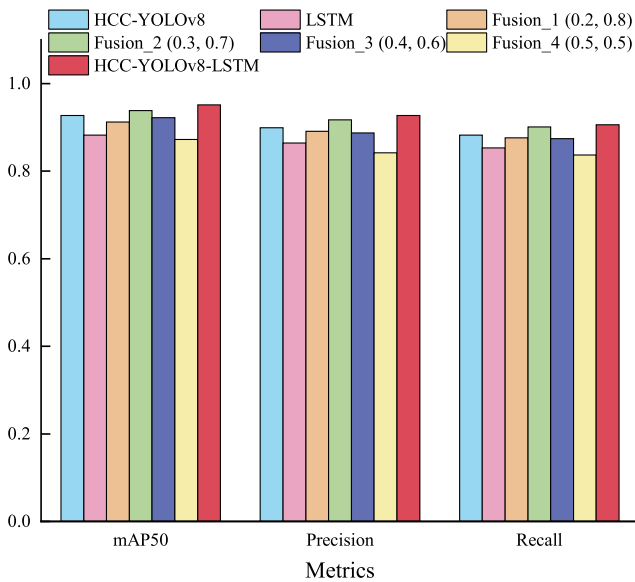


Fig. 8. Comparative evaluation of different fusion strategies.

the predicted and actual feature points.

3.3. Real scenario evaluation

With the development of smart agriculture, the application of computer vision in animal husbandry has become increasingly widespread. Evaluating a model's performance and robustness in real-world scenarios has become an important step. We conducted a series of tests on the HCC-YOLOv8-LSTM model in a real pig farm environment.

At two partner pig farms in Shanxi Province, we collected data on the excretion behavior and feces of pigs suffering from various diseases under different lighting conditions using Hikvision DS-2CD 3345 D-I cameras. Additionally, temperature and humidity sensors on the pig farms were used to record the average temperature and humidity each day, as shown in Table 5. To ensure the reliability of the data, we employed domain experts for annotation.

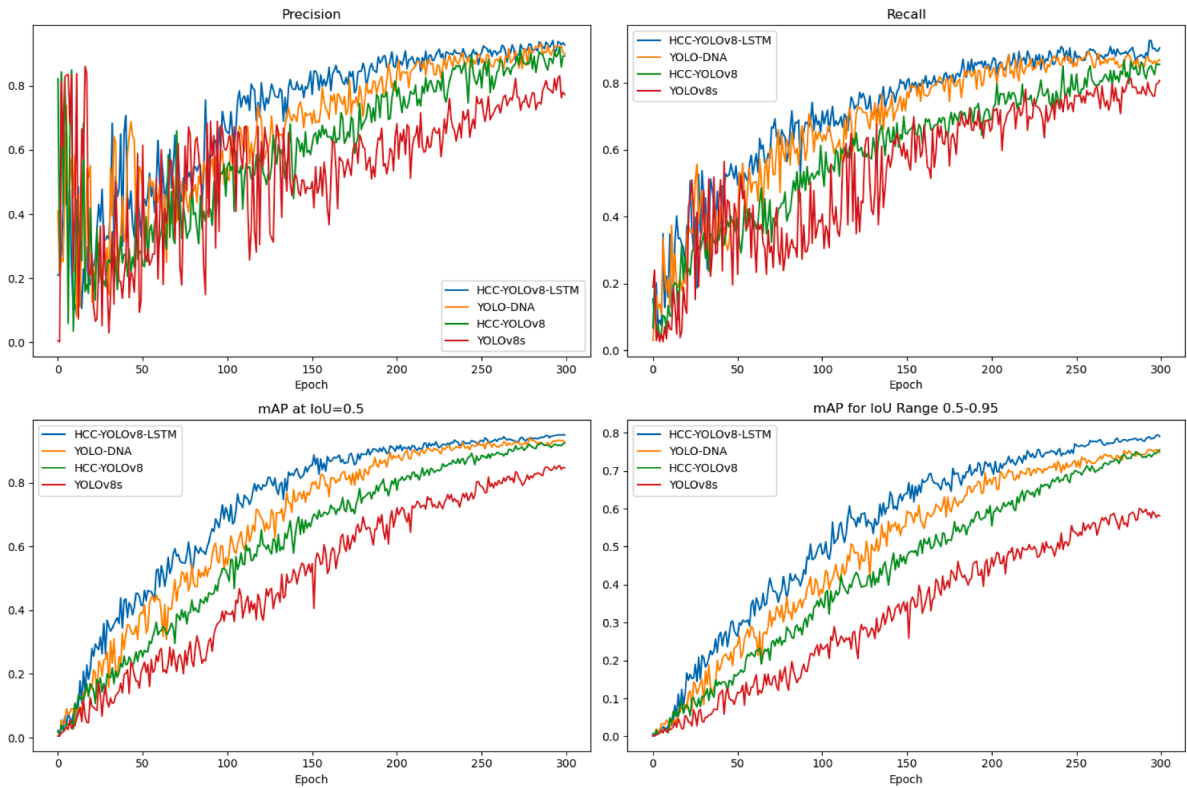
We collected defecation behavior and feces data from pigs at different growth stages, breeds, and lighting conditions across various cooperative pig farms. This diversity in data sources introduced challenges that could impact model performance and thoroughly validated the model's generalizability. In addition, we collaborated with veterinary experts to perform real-time quantitative PCR (qPCR) tests on pig feces, further strengthening the validation of the model's reliability. As shown in Table 6, the comparison between the model's predictions and the pigs' medical test results yielded satisfactory outcomes, confirming that the model's performance is closely aligned with expert evaluations. These real-world evaluation results provide proof of the model's effectiveness in actual pig farming environments, aiding in the optimization and application of the model. We believe that with ongoing improvements and testing, HCC-YOLOv8-LSTM will unleash significant potential in the agricultural sector.

4. Discussions

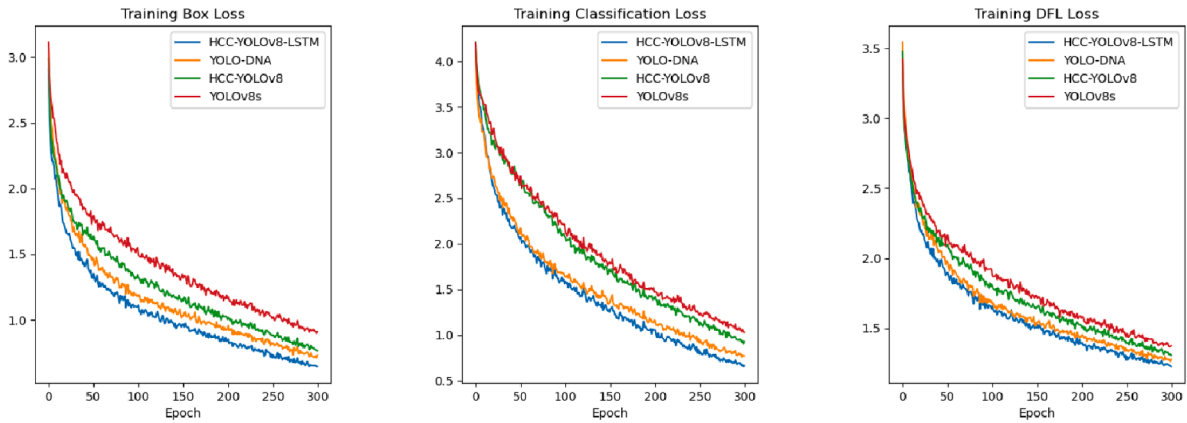
4.1. Model performances of unimodal and multimodal models in pig disease recognition

In the context of pig disease detection, the initial use of the HCC-YOLOv8 model for disease diagnosis based solely on features extracted from pig feces resulted in an accuracy of 0.899 and an mAP50 of 0.927. This indicates significant misclassifications when only excrement features are used for disease identification. Several factors could explain this, including external environmental conditions such as inadequate lighting, which may mask fecal characteristics, especially for diseases such as TGE and SD where feces are light in color. Additionally, pigs walking around and stepping on excrement can obscure their features, and issues such as diarrhea from unclean water can further complicate feature clarity.

To address these issues, we integrated abnormal excretion behaviors with excrement information and used the HCC-YOLOv8-LSTM model for pig health monitoring. This model combines both excretion behaviors and excrement features, reducing misclassifications caused by external interference. As shown in Table 7, the HCC-YOLOv8-LSTM model performs better. Compared with the multimodal model YOLO-DNA, our model has greater accuracy with an mAP50 of 0.951. It also shows improvements over HCC-YOLOv8, with precision increasing by 0.028 and



(a) Performance



(b) Loss

Fig. 9. Comparison results between multimodal and unimodal models.

Table 5
Data collection details from cooperative pig farms in Shanxi province.

Date	Location	Breed	Age	Date Volume	Lighting Condition	Average temperature	Average humidity
2024-06-18	Fenyang	Long White Pig	Fattening Pigs	Excretion: 300 Manure: 460	Bright	30.2°C	64.7 %
2024-06-19	Fenyang	Long White Pig	Fattening Pigs	Excretion: 120 Manure: 180	Dark	29.4°C	65.3 %
2024-06-27	Puxian	Large White Pig	Nursery Pigs	Excretion: 320 Manure: 470	Bright	31.3°C	67.8 %

Table 6

Comparison of model predictions with domain expert assessments.

Indicators	Model Prediction	Expert Assessment	Difference
mAP50	0.951	0.966	-0.018
Recall	0.906	0.899	+0.007
F1-score	0.917	0.921	-0.004

Table 7

Performance comparison of various YOLOv8s models.

Type	Model	Precision	mAP50	mAP
Unimodal	YOLOv8s	0.861	0.887	0.714
	HCC-YOLOv8	0.899	0.927	0.748
Multimodal	HCC-YOLOv8-LSTM	0.927	0.951	0.790
	YOLO-DNA	0.913	0.940	0.766

mAP50 increasing by 0.024, and over the baseline YOLOv8s, with precision increasing by 0.066 and mAP50 by 0.064. This confirms the effectiveness of our fusion model for accurate identification in the complex environment of pig farms, and its lightweight design allows for effective deployment in such settings.

Overall, our model has made significant progress in pig disease recognition, achieving substantial improvements in several metrics, including mAP50, mAP, precision, recall, and the F1 score. The integration of multimodal information has made disease diagnosis more comprehensive and accurate. This highlights the potential of the model for early disease detection in intelligent pig farming environments, where timely and accurate disease identification is critical for preventing large-scale outbreaks. Additionally, the lightweight design of our model is particularly suitable for real-world deployment in pig farms. Given the resource constraints in agricultural environments, computational efficiency and speed are essential. The improvements in model performance, along with its adaptability to real-time monitoring, make it a viable solution for non-invasive and intelligent early disease prevention in the pig farming industry.

4.2. Limitations of the proposed model

Although the HCC-YOLOv8-LSTM model demonstrates strong performance in detecting pig excretion behavior and feces and its lightweight design makes it suitable for real-world farm deployment, several limitations still need to be addressed. External environmental factors can impact model performance, especially in cases of insufficient or excessive lighting, where the morphological features of feces can easily be misjudged, affecting the detection accuracy of the model. The diversity and generalizability of the dataset need further enhancement. Although we achieved good results on the current dataset, the model's generalizability has not yet been validated across a broader range of pig farm environments. The varying excretion behavior patterns and feeding methods of different pig farms may result in the model's performance in new scenarios falling short of expectations. The impact of different feed types on the model's detection results is another factor worth considering. The type and composition of feed not only affect pig growth but can also lead to significant changes in the morphology and composition of excrement, presenting new challenges for model recognition accuracy.

Addressing these limitations is critical for the successful application of this model in real pig farming environments. Thorough field testing and corresponding model iterations are essential.

5. Conclusion

The focus of this study was to develop a pig disease monitoring approach by integrating pig feces and excretion behavior analysis through a fusion recognition model. The HCC-YOLOv8-LSTM model

achieved notable improvements in precision, mAP, recall, parameter efficiency, and GFLOPs, making it suitable for real-world applications because it maintains a lightweight design optimized for pig farm environments.

This study presents an effective strategy for early disease detection in pigs, surpassing the performance of other existing models. Future efforts will concentrate on further refining and optimizing the HCC-YOLOv8-LSTM model to improve the accuracy of monitoring various pig diseases. The study will also expand to include datasets collected under different lighting conditions, camera angles, and environmental variables to assess the model's generalization capabilities. The lightweight nature of the model allows deployment on embedded or mobile devices, facilitating real-time disease warnings on pig farms. Moreover, field implementation in digital smart farming setups will be explored to enhance the model's practical utility in reducing disease-related losses. Overall, the HCC-YOLOv8-LSTM model has strong potential for advancing smart livestock farming and offers valuable technical support for the intelligent pig farming industry.

CRedit authorship contribution statement

Haopu Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Bugao Li:** Resources, Funding acquisition. **Haoming Li:** Data curation. **Min Chen:** Data curation. **Yanbo Song:** Project administration. **Zhenyu Liu:** Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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